LMA INDIVIDUAL PROJECT

PHASE 1 — DATA COLLECTION AND TOKENIZER CONSTRUCTION

Hindi + Nepali | End-to-End Final Report

Item	Phase 1 configuration
Higher-resource language (Model H)	Hindi
Lower-resource language (Model L)	Nepali
Tokenizer	Byte Pair Encoding (BPE), trained from scratch
Final tokenizer vocabulary	31,744 per language
BPE merges	31,488 per language
Final training corpus target	≈500M tokens per language
Manual training-data target	≥20% per language
Primary vocabulary experiment	8,192 → 65,536
Evaluation documents	10,000 per language
Phase 1 status	Complete

This updated report incorporates the completed corpus split verification, final artifact inventory, manual-vs-downloaded provenance, tokenizer experiments, stability analysis, and repository submission checklist.

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1. Project Requirements and Phase 1 Scope

Phase 1 requires two independent monolingual dataset and tokenizer pipelines: one for a higher-resource non-English Indian language and one for a lower-resource language. The target is approximately 500M training tokens per language after tokenization. If the lower-resource language cannot reach the target, the maximum feasible amount and exact shortfall must be reported.

At least 20% of the final training tokens must come from manual collection for both languages. Documents must not be shared between the two corpora, and each language must maintain independent train/validation/test splits and statistics.

For tokenization, each language requires its own BPE tokenizer trained from scratch. Pretrained tokenizers and shared vocabularies are not permitted. Phase 1 also requires vocabulary-size evaluation, fertility, characters-per-token analysis, tokenization examples, unknown-token statistics, tokenizer model/vocabulary artifacts, and reproducible code.

2. Language Selection

2.1 Hindi — Model H

Hindi was selected as Model H because it is a relatively high-resource Indian language with substantial public corpus availability. The supplied Hindi Raw Text Corpus documentation describes Hindi as an Indo-Aryan language written in Devanagari and reports collection across libraries and other sources, with literary and non-literary material.

2.2 Nepali — Model L

Nepali was selected as Model L because it is included in the assignment's allowed lower-resource language list. The supplied Nepali corpus documentation describes collection from libraries in Darjeeling, Sikkim, Assam and Uttaranchal and highlights greater difficulty obtaining scientific, technical, finance and similar domains. Orthographic variation in Devanagari makes normalization particularly important.

3. Corpus Design

Dimension	Hindi	Nepali
Language tier	Higher-resource (Model H)	Lower-resource (Model L)
Corpus type	Monolingual	Monolingual
Primary public/non-manual source	IndicCorp	Downloaded/non-manual corpus
Manual sources in final construction	Manual corpus component	Nepalipatra + KhabarHub and other manual sources
Vocabulary shared?	No	No
Tokenizer shared?	No	No
Final token target	499,999,998	499,999,956
Manual tokens	100,000,000	99,999,956
Manual fraction	20.00%	20.00%
Non-manual tokens	399,999,998	400,000,000

4. Data Collection — Hindi

IndicCorp was the major high-volume non-manual Hindi source used for the final incremental tokenization stage. The project execution record confirms 70 cleaned IndicCorp chunks. An earlier Hindi preprocessing checkpoint recorded 13,930,585 records seen, 6,965,293 records saved, and 400,000,021 proxy words. A previously tokenized Hindi checkpoint contained 240,262 documents, 27,552,606 words, 156,640,899 characters and 67,722,646 tokens.

The wider Hindi project source inventory also includes AllenAI C4, Wikimedia/Wikipedia and a Hindi web-crawl batch. Manual Hindi sources include Aaj Tak, Hindi news material, KavitaKosh, Hindi RSS material, manually collected Hindi Wikipedia, Wikisource and additional manually collected material.

5. Data Collection — Nepali

The Nepali pipeline combines downloaded/non-manual material with a substantial manually collected component. The final manual incremental-tokenization stage explicitly used Nepalipatra and KhabarHub.

5.1 Nepalipatra

Statistic	Value
Documents	14,363
Words	2,623,251
Characters	17,216,800
Bytes	46,041,328
Files	3

The three Nepalipatra chunks contain 5,000, 5,000 and 4,363 documents.

5.2 KhabarHub

The KhabarHub manual collection contains 23 article chunks. The final incremental tokenization stage processed these together with the three Nepalipatra chunks.

5.3 Other Nepali Manual Sources

- Kantipur — manually extracted web material.
- Sahityaras — manually extracted literary material.
- CDC — manually extracted material from PDFs.
- Internet Archive OCR — OCR-derived Nepali material.
- National Library PDF extraction — manually extracted/processed PDF material.
- Additional manually extracted web material and related shards.

6. Cleaning, Normalization and Corpus Quality

- Unicode text is retained as the underlying corpus representation.
- Invalid and unwanted material is removed during cleaning.
- Foreign-script contamination is filtered where identified.
- Duplicate and invalid records are removed by the preprocessing pipeline.
- Language-specific orthographic issues are considered, particularly for Nepali.
- Hindi source documentation records validation, normalization and proofreading.
- Hindi and Nepali remain in independent directory structures and processing pipelines.

The cleaning stage is therefore more than raw-text concatenation: collection, filtering, normalization, deduplication and language-specific quality checks are performed before final corpus construction.

7. Manual-Data Construction and the 20% Requirement

Language	Manual tokens	Non-manual tokens	Final total	Manual %
Hindi	100,000,000	399,999,998	499,999,998	20.00%
Nepali	99,999,956	400,000,000	499,999,956	20.00%

Both languages satisfy the ≥20% manual-data requirement. The final totals are slightly below exactly 500M because incremental tokenization respects complete document boundaries rather than truncating documents.

8. Initial Tokenizer Training

Tokenizer property	Hindi	Nepali
Algorithm	BPE	BPE
Vocabulary	31,744	31,744
BPE merges	31,488	31,488
Training	From scratch	From scratch
Shared vocabulary	No	No
Tokenizer file	hindi_bpe_31744.json	nepali_bpe_31744.json

The tokenizers are language-specific and independently trained. No pretrained tokenizer or shared vocabulary was used.

9. Tokenizer Sanity Tests

Test	Hindi	Nepali
Encoded token count in sanity test	17	13
Round-trip reconstruction	True	True

The sanity tests confirmed successful encode/decode round-trip reconstruction for both tokenizers.

10. Final Incremental Tokenization

10.1 Hindi

Hindi incremental tokenization began with 67,722,646 existing non-manual tokens and targeted 332,277,354 additional tokens. The final run processed 70 IndicCorp chunks.

Hindi token budget	Tokens
Existing non-manual	67,722,646
Additional IndicCorp	332,277,352
Final non-manual	399,999,998
Manual component	100,000,000
Final total	499,999,998

The run generated 332,277,352 additional tokens and stopped two tokens short of the requested incremental target because the next complete document would have exceeded the target. It processed 2,643,291 documents, 151,813,130 words and 774,078,067 characters.

10.2 Nepali

Nepali incremental tokenization targeted 68,806,066 additional tokens on top of an existing 31,193,934-token manual pool. The run processed 26 files: 3 Nepalipatra and 23 KhabarHub files.

Nepali token budget	Tokens
Existing manual	31,193,934
Additional manual	68,806,022
Final manual	99,999,956
Non-manual	400,000,000
Final total	499,999,956

The run generated 68,806,022 additional tokens and stopped 44 tokens short because the next complete document would have exceeded the target. It processed 119,022 documents, 30,432,235 words and 200,620,657 characters.

11. Vocabulary-Size Experiment

A controlled tokenizer-selection experiment used fixed training corpora and a fixed evaluation set. The training corpora contained 80,104,568 Hindi characters and 80,065,007 Nepali characters. Evaluation contained 10,000 documents per language.

Vocabulary	Hindi fertility	Nepali fertility	Hindi chars/token	Nepali chars/token
8,192	2.7328	2.6989	2.0622	2.1996
16,384	2.5593	2.5446	2.2020	2.3330
24,576	2.4856	2.4801	2.2673	2.3937
32,768	2.4434	2.4417	2.3064	2.4313
49,152	2.3964	2.3975	2.3517	2.4761
65,536	2.3705	2.3717	2.3774	2.5031

Fertility decreases monotonically as vocabulary size increases for both languages. The largest early gain is between 8,192 and 16,384, after which the marginal improvement becomes smaller.

12. Fertility Stability Experiment

To test whether the fertility trend depended on one evaluation sample, the 10,000-document evaluation set was divided into five fixed subsets. Each subset contained 2,000 Hindi documents and 2,000 Nepali documents.

Vocab	Hindi mean	Hindi std	Nepali mean	Nepali std
8,192	1.961455	0.016113	1.582750	0.066696
16,384	1.630386	0.013444	1.405148	0.076795
24,576	1.516787	0.012025	1.332388	0.077329
32,768	1.456364	0.010704	1.290962	0.075995
49,152	1.389938	0.009890	1.244039	0.072587
65,536	1.353215	0.009274	1.217993	0.069338

Hindi shows low between-subset variation, with standard deviation decreasing from 0.016113 at 8K to 0.009274 at 65K. Nepali has higher variation between subsets, but its standard deviation is also lower at 65K than at 16K and remains consistent with the same decreasing fertility trend.

Important metric distinction: the absolute fertility values in the stability experiment are not numerically interchangeable with the primary vocabulary experiment because the experiments use different evaluation

formulations. The primary experiment is the main vocabulary-size comparison; the stability experiment tests consistency across fixed subsets.

13. Interpretation of Fertility and Characters per Token

BPE begins with smaller units and learns frequent adjacent symbol sequences as merges. With a larger vocabulary, more frequent multi-character and multi-subword sequences can be represented directly. Fewer tokens are therefore needed to represent the same text, reducing fertility and increasing average characters per token.

The experiment demonstrates this empirically: both Hindi and Nepali show lower fertility at every successively larger vocabulary size, while characters per token increase from 8K through 65K.

14. Final Vocabulary Choice

The final tokenizer vocabulary is 31,744 entries per language with 31,488 BPE merges. This is close to the 32,768 point explored in the vocabulary experiment, where the major early fertility gains have already been captured while avoiding the progressively larger vocabulary sizes. It is therefore a practical compromise between tokenization efficiency and vocabulary size.

15. Final Train/Validation/Test Split Verification

The final Phase 1 split stage was completed and verified after the earlier splitting run. The following values are the recorded verification outputs. The split statistics report document and character counts; they should not be interpreted as token counts.

15.1 Hindi Split Verification

Split	Documents	Characters
Train	7,061,250	2,152,245,131
Validation	71,912	21,537,503
Test	72,393	22,152,082

Hindi split verification completed successfully.

15.2 Nepali Split Construction and Verification

The Nepali final split stage read the three manual-tokenization input files and found 45,621 valid records. The resulting 80/10/10 split was written as gzip-compressed JSONL.

Split	Documents	Characters
Train	36,496	55,164,938
Validation	4,562	6,685,307
Test	4,563	6,979,674

Nepali split artifact	Path
Train	nepali/splits/train/nepali_train_00000.jsonl.gz
Validation	nepali/splits/validation/nepali_validation_00000.jsonl.gz
Test	nepali/splits/test/nepali_test_00000.jsonl.gz
Split statistics	phase1_report/nepali_split_statistics.json

The Nepali split totals are 45,621 documents: 36,496 train, 4,562 validation and 4,563 test. The split stage therefore completed with explicit, reproducible statistics.

16. Phase 1 Artifact Inventory

Artifact	Purpose
hindi_bpe_31744.json	Hindi trained BPE tokenizer/model
nepali_bpe_31744.json	Nepali trained BPE tokenizer/model
vocabulary_fertility_results.csv	Primary vocabulary-size fertility results
vocabulary_fertility_results.json	Primary fertility results in JSON
fertility_vs_vocab_size.png	Primary fertility graph
fertility_stability_results.csv	Five-subset fertility stability results
fertility_stability_results.json	Stability results in JSON
fertility_stability_vs_vocab_size.png	Stability graph
fertility_comparison.png	Hindi/Nepali fertility comparison graph
corpus_composition.png	Corpus composition visualization
manual_data_percentage.png	Manual-data percentage visualization
dataset_statistics.csv	Dataset-level statistics
phase1_metrics.csv	Consolidated Phase 1 metrics
phase1_metrics.json	Consolidated Phase 1 metrics in JSON
split_statistics.json	Final split verification statistics
nepali_split_statistics.json	Nepali split statistics
tokenization_examples.md	Tokenization examples
phase1_report.md	Repository markdown report

The repository also contains the language-specific data collection, preprocessing, split, tokenizer training, evaluation and supporting scripts. Large corpus data remains in external storage as required; the repository stores the reproducibility code, statistics, model/tokenizer artifacts and graded figures.

17. Assignment Deliverables Mapping

Assignment requirement	Phase 1 status / artifact
Dataset collection scripts	Completed — Hindi and Nepali language-specific collection pipelines
Dataset preprocessing	Completed — independent cleaning/processing paths
Per-language dataset statistics	Completed — dataset and corpus statistics artifacts
Train/validation/test splits	Completed and verified — split statistics and compressed split files
Tokenizer training code	Completed — separate BPE implementations
Vocabulary/model files	Completed — 31,744-vocabulary Hindi and Nepali tokenizer files
Held-out vocabulary evaluation	Completed — 8,192 to 65,536 fertility experiment
Stability analysis	Completed — five fixed evaluation subsets
Tokenization examples	Completed — tokenization_examples.md
Graphs / graded figures	Completed — fertility, stability and corpus-composition figures
Phase 1 report	Completed — this DOCX and phase1_report.md

18. Final Phase 1 Results

Final result	Hindi	Nepali
Final training tokens	499,999,998	499,999,956
Manual tokens	100,000,000	99,999,956
Manual fraction	20.00%	20.00%

Non-manual tokens	399,999,998	400,000,000
BPE vocabulary	31,744	31,744
BPE merges	31,488	31,488
Sanity round-trip	True	True

Both final corpora are approximately 500M tokens and meet the manual-data requirement. The final tokenizers are independently trained, language-specific and validated by round-trip reconstruction.

19. Repository / GitHub Submission Checklist

- ✓ Hindi collection and preprocessing code committed.
- ✓ Nepali collection and preprocessing code committed.
- ✓ Both tokenizer training implementations committed.
- ✓ Hindi and Nepali tokenizer/model files committed or linked where size requires external storage.
- ✓ Final dataset statistics committed.
- ✓ Final train/validation/test split statistics and split artifacts committed/linked as required.
- ✓ Vocabulary fertility CSV/JSON results committed.
- ✓ Vocabulary fertility graph committed.
- ✓ Fertility stability CSV/JSON results committed.
- ✓ Fertility stability graph committed.
- ✓ Corpus composition and manual-data percentage figures committed.
- ✓ Tokenization examples committed.
- ✓ phase1_report.md committed under the Phase 1 reporting area.
- ✓ Large datasets retained in required external storage with accessible README documentation.
- ✓ Drive/external-storage documentation identifies where raw, processed, split and report artifacts are located.
- ✓ Phase 1 branch contains the graded figures/tables so graders do not need to retrieve them from Drive.

20. Reproducibility and Storage Notes

Final corpus construction stops before a complete document would exceed the requested exact token target. This explains the small deviations from exactly 500M tokens: Hindi finishes at 499,999,998 and Nepali at 499,999,956. These values arise from respecting document boundaries rather than arbitrarily truncating text.

The vocabulary experiments use a separate fixed 80M-character training/evaluation setup. They are tokenizer-selection experiments and should not be confused with the final approximately 500M-token training corpora.

Large raw/processed corpora are not expected to be stored directly in GitHub. The repository should contain code, statistics, tokenizer/model artifacts, reports and graded figures, while Drive/external storage should contain the large corpus files. The README/Phase 1 documentation should provide the corresponding storage paths.

21. Source Inventory — Manual vs Downloaded / Extracted

21.1 Hindi — Downloaded / Non-manual

- IndicCorp — primary non-manual source used for the final Hindi incremental expansion; 70 cleaned chunks.
- AllenAI C4 — downloaded Hindi source family recorded in the project.
- Wikimedia/Wikipedia Hindi — downloaded source family recorded in the project.
- Hindi web-crawl batch — downloaded/raw source family recorded in the project.

21.2 Hindi — Manual / Extracted

- Aaj Tak — manual news extraction.
- Hindi news material.
- KavitaKosh.
- Hindi RSS extra material.
- Manually collected Hindi Wikipedia material.
- Hindi Wikisource.
- Additional manually collected material.

21.3 Nepali — Downloaded / Non-manual

- nepali_train.jsonl — downloaded Nepali training corpus.
- nepali_test.jsonl — downloaded Nepali test corpus.

21.4 Nepali — Manual / Extracted

- Nepalipatra — manually collected news articles; 14,363 articles and 2,623,251 words in the recorded source statistics.
- KhabarHub — manually collected news articles; final incremental tokenization used 23 chunks.
- Kantipur — manually extracted web material.
- Sahityaras — manually extracted literary material.
- CDC — manually extracted PDF material.
- Internet Archive OCR — OCR-derived material.
- National Library PDF extraction.
- Additional manually extracted web material.

21.5 Sources Used in the Final Incremental Runs

Language	Final incremental source(s)	Role
Hindi	IndicCorp — 70 cleaned chunks	Non-manual expansion
Nepali	Nepalipatra — 3 chunks + KhabarHub — 23 chunks	Manual expansion

The complete raw-source inventory is larger than the exact final token budget. Therefore, the report distinguishes the broader source pool from the sources explicitly consumed in the final incremental runs.

22. Source Documents and Evidence

- LMA Individual Project specification — requirements for corpus construction, manual data, tokenizer training and deliverables.
- Hindi Raw Text Corpus documentation — Hindi source provenance, domains, validation, normalization and proofreading.
- Nepali Raw Text Corpus documentation — Nepali source provenance, field collection, domain coverage and orthographic issues.
- Phase 1 execution logs — project paths, tokenizer models, incremental tokenization, source counts and final corpus summary.
- Nepali source statistics — Nepalipatra/KhabarHub document, word, character and file statistics.
- Vocabulary fertility experiment outputs — vocabulary_fertility_results.csv/json and fertility_vs_vocab_size.png.
- Fertility stability outputs — fertility_stability_results.csv/json and fertility_stability_vs_vocab_size.png.
- Final split verification outputs — split_statistics.json and nepali_split_statistics.json.

The source-provenance section records both the source families documented in the project and the specific sources used during final incremental tokenization. It does not imply that every raw source contributed equally to the final 500M-token budgets.

23. Phase 1 Conclusion

- Hindi and Nepali were maintained as fully independent monolingual pipelines.
- Hindi was used as the higher-resource language and Nepali as the lower-resource language.
- The final training corpora contain 499,999,998 Hindi tokens and 499,999,956 Nepali tokens.
- Both languages contain 20.00% manual data in the final corpus summary.
- Separate BPE tokenizers were trained from scratch with 31,744 vocabulary entries and 31,488 merges per language.
- Tokenizer round-trip sanity tests passed for both languages.
- The vocabulary-size experiment shows decreasing fertility and increasing characters per token as vocabulary size grows.
- The five-subset stability experiment confirms that the decreasing fertility trend is not dependent on a single evaluation sample.
- Final train/validation/test split construction and verification were completed, with split statistics saved as explicit artifacts.
- Required Phase 1 statistics, graphs, tokenizer artifacts, examples and report artifacts were prepared for repository submission.
- Phase 1 is complete and provides the data/tokenizer foundation for Phase 2 model work.

Appendix A — Key Project Paths

Area	Repository / storage location
Hindi pipeline	hindi/data/ and hindi/tokenizer/
Nepali pipeline	nepali/data/ and nepali/tokenizer/
Hindi tokenizer	hindi/model/hindi_bpe_31744.json
Nepali tokenizer	nepali/model/nepali_bpe_31744.json
Phase 1 report	reports/phase1_report.md
Pipeline notebook	scripts/phase1_pipeline.ipynb
Bootstrap script	scripts/bootstrap_colab.py
Manual scraper	scripts/scrape_manual.py
Nepali split train	nepali/splits/train/nepali_train_00000.jsonl.gz
Nepali split validation	nepali/splits/validation/nepali_validation_00000.jsonl.gz
Nepali split test	nepali/splits/test/nepali_test_00000.jsonl.gz
Large corpus storage	External Drive / project storage; linked from README

END OF PHASE 1 REPORT